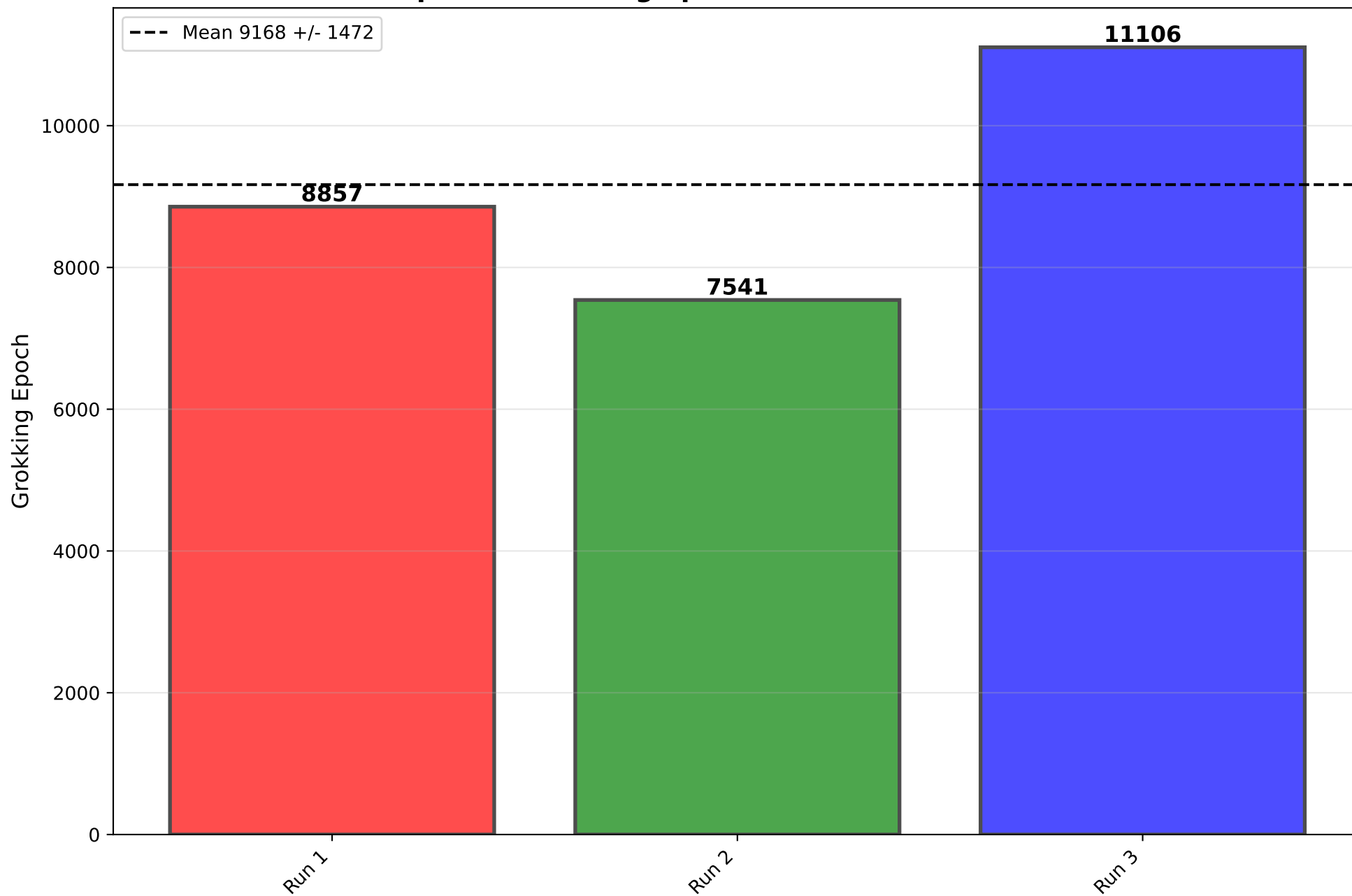
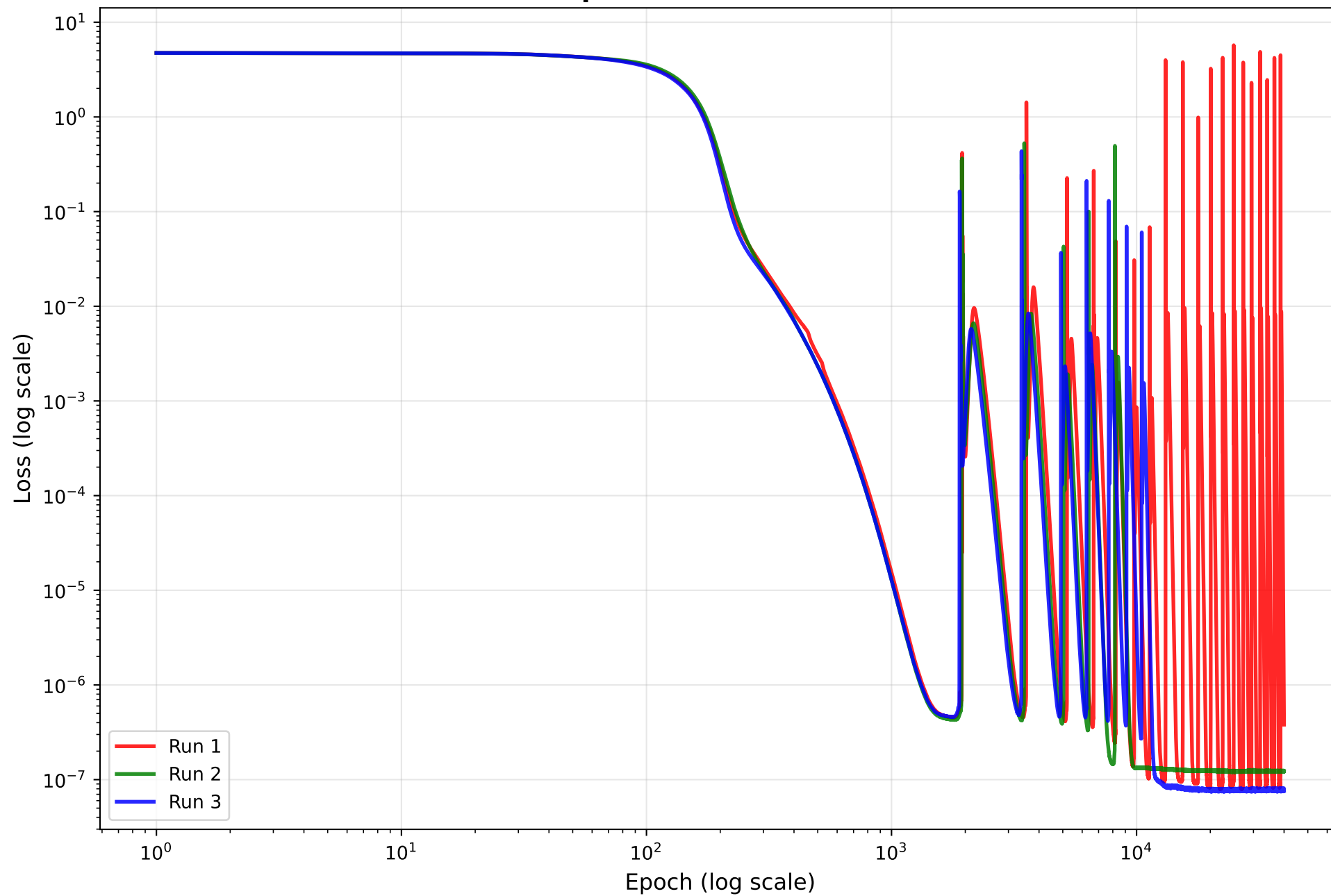


p=113 Grokking Epoch Across Seeded Runs



p=113 Loss Curves



p=113 RUN CONSISTENCY ANALYSIS (L2-Norm predictor)

Grokking Epochs (3 seeded runs):

Run 1: 8857 epochs

Run 2: 7541 epochs

Run 3: 11106 epochs

Mean: 9168.0 +/- 1471.9 epochs

Consistency: Low

Setup:

Task : (a + b) mod 113

Optimizer : AdamW lr=1e-3 wd=1.0 betas=(0.9, 0.98)

Model : TransformerFourHead, 4 heads, d_model=128

Predictor : L2 Norm only

Interpretation:

- Low std dev: predictor signal consistent across random seeds
- High std dev: predictor signal sensitive to initialization

Next Steps:

1. Overlay the L2 norm curves on Nanda Figure 7 (Literature/nanda_figures/)
2. Check whether betas=(0.9, 0.98) shrank the pre-grok weight-norm spike
3. Validate the L2 Norm predictor against the 3-criteria protocol